

Police Gazette Transcripts (PDF to Excel)

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1 Introduction

This document describes our approach to extract the tables from the Police Gazette files (in .pdf format), and store the data into Excel files.

The Gazette files contain two different types of tables, with information about the convicted and discharged individuals. Since the number of these tables is relatively large, we were particularly interested in how to store and organize data for easy access, and updates. Our solution is presented in Section 2.

Section 3 contains the technical details of how to extract the tables from the Gazette files and store them into Excel files.

2 The Design of the Transcript

2.1 Storing one table

We propose storing **each table** we extract from the Gazette files in **one** single Excel file. The naming conventions for the Excel files is described in Section 2.2.

The advantages of storing each table in one single file are listed below:

1. Extracting different tables, as well as checking the correctness of the extracted tables, can be done independently, by different people.
2. Checking for correctness of data is much easier this way, as each table contains an easily-manageable number of records (e.g. no more than 30 records).
3. It is easy to keep track of the work done, as we will show below.
4. It is easy to merge all these separate files into a single file that contains all the records.

2.2 Naming conventions

Assume we extract a table from file `AU6103C Tasmania Police XXXX-YYYY.pdf`, where `XXXX-YYYY` is the time period this file contains info about. Assume that this table is located on page `P`.

We propose the following naming conventions:

1. The name of the Excel file that contains the table above is `AU6103C-XXXX-YYYY-P.xls`.
2. If there are more tables on page `P`, and the table we extract is the `i`-th table on the page, the name of the file containing it is `AU6103C-XXXX-YYYY-P-i.xls`. Consequently, for a page with `n` tables, $i \in 1 \dots n$.

Note: All the file names in the Police Gazette collection contain the prefix “AU6103C”. To make it easier to locate the Excel files, and, consequently, the data they contain, we have decided to keep the time period in the name of the file.

2.3 Design of the metatable

We propose having a metatable, stored as a table in an Excel file (`police_gazette_metable.xls`), that will contain meta-information about all the tables we extract from the Gazette files.

Each **row** in this metatable contains meta-information about one single file storing one single table (as described in Section 2.1).

Meta-information pertaining to one Excel file is stored only once in this metatable, rather than for each record in the Excel file. For instance, the file `AU6103C-XXXX-YYYY-P.xls` contains 23 records. Rather than having a column `PlaceOfConviction` in the Excel file, with the value “Hobart Town” for all 23 records, we can have the `PlaceOfConviction` column in the metatable, with the value “Hobart Town”, and it is stored only once.

The header of the metatable describes the meta-information we store, and consists of, but is not limited to, the columns (referred to as “fields”) below:

1. **Index:** this is a numeric field, and its value starts at 1.
This index has no meaning with respect to the data, but will be useful for us. For instance, it’s easier to refer to a record by using this index, rather than the name of the file it describes (filename column below).
2. **Filename:** this is the name of the file, and is stored as a hyperlink to the file with this name.
Making this column a hyperlink has the advantage of easy access to the file. A file can be opened with a single click. Section 4.1 contains a short description of how to insert hyperlinks in Excel.
3. **Description:** this is a text field, and contains the information above the table in the `.pdf` file.

Note: this information contains the time and the place of when and where the decision was made. We can easily extract the time and place later into new columns in the metatable.

- In the “Discharged” tables we have 2 dates listed, and I am not sure which one is needed; you can add this info in the example below.
- In the “Discharged” tables we have information about the House of Correction.
- In the “Convicted” tables we have information about the Supreme Court.

Based on my understanding, we can have:

- (a) **Place** – this can store the place of “House Of Correction” for “Discharged”, and “Supreme Court” for “Convicted”.
- (b) **DateHouseOfCorrection** or **DateCountryDistrict**, or both, for “Discharged” and **DateSupremeCourt** for “Convicted”, **OR** **Date** for both “Discharged” and “Convicted”, assuming we need only one of the dates for “Discharged”.

4. **Type:** this is a text field, containing one of the values **Discharged**, **Convicted**, according to the information stored in the file.
This information is also captured in the **Description** field, but having it in a separate column will be very helpful.
5. **SignedBy:** this is a text field, and contains the information below the table in the **.pdf** file.
6. **RowCount:** this is an optional numeric field, and represents the number of records in a file. It is intended for verification purposes only.
7. **ExtractedBy:** this is an optional text field, and contains the name of the person who extracted the file.
8. **CheckedBy:** this is an optional text field, and contains the name of the person who checked the correctness of the file.

Below we present an example of a record in the metatable. Assume we store the meta-information for file AU6103C-XXXX-YYYY-P.xls. It will be stored as one record in the metatable, with the following values:

1. **Index:** N;
2. **Filename:** AU6103C-XXXX-YYYY-P.xls;
3. **Type:** Discharged;
4. **Description:** *PRISONERS discharged from H.M. GAOLS and HOUSES of CORRECTION,-Hobart Town and Launceston, during the week ending 17th January ; Country Districts, for the Week ending 13th January, 11166.;*
[Please fill in the info below, as needed]
5. **Date** (maybe? TBD);
6. **Time** (maybe? TBD);
7. **SignedBy:** *J. FORSTER, Inspector. JAMES BARNARD, Goserrment Printer, Hobart Town.;*
8. **RowCount:** 23;
9. **ExtractedBy:** Ramona;
10. **CheckedBy:** TBD.

3 Extracting data from Police Gazette files

The software we use to extract and convert data from pdf to Excel format is PDF2XL, available at <http://www.cogniview.com/pdf2xl.php>. This task is accomplished by following the steps below:

1. **Open** (File|Open) the **.pdf** file;
2. **Open Layout** (Layout|Open Layout) and choose one of the `police_gazette_discharged.layout`¹, or `police_gazette_convicted.layout`² files, both provided by us.
 - **Note:** these are the layouts we have defined, and they will assure uniformity.
 - (a) In some cases, because of millimetric variations between tables, some manual manipulation of the column separators is needed. When the cursor is moved over these separators, a double-pointing arrow will appear; by dragging it, the separator can be moved.

¹This layout is for tables containing info about discharged individuals.

²This layout is for tables containing info about convicted individuals.

- (b) **Edit|Enable Row and Cell Editing** allows to adjust the row size. This is necessary for the records that expand on more than one row. Some very useful options are **Merge** and **Split Here**, available on right-click.
- 3. **Convert Excel (Convert|Excel)** will convert the file. We will use the naming conventions described in Section 2.2, and we will convert one page at a time.

4 Appendix

4.1 Hyperlink in Excel

Use the following steps to insert a hyperlink in Excel:

1. Position the cursor on the cell you want to insert the hyperlink.
2. **Insert|Hyperlink**. A dialog box will ask for the path of the file, and other information.